

Expense Tax Elasticity and the Cross Section of Stock Returns

I. M. Harking

October 21, 2025

Abstract

This paper studies the asset pricing implications of Expense Tax Elasticity (ETE), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on ETE achieves an annualized gross (net) Sharpe ratio of 0.40 (0.34), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 40 (31) bps/month with a t-statistic of 4.95 (3.81), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in current operating assets, Employment growth, Asset growth, Long-term EPS forecast, change in ppe and inv/assets, Change in current operating liabilities) is 24 bps/month with a t-statistic of 2.84.

1 Introduction

Financial markets process vast amounts of information daily, yet mounting evidence suggests that prices do not instantly reflect all available information as predicted by the efficient market hypothesis. Instead, information diffuses gradually through markets, creating predictable patterns in asset returns as investors slowly incorporate new signals into prices. While extensive research documents various market frictions that impede information processing—from limited attention to cognitive constraints—we still lack a complete understanding of which specific accounting signals experience the slowest diffusion and generate the most persistent return predictability. This paper examines a novel signal, Expense Tax Elasticity (ETE), which captures firms’ sensitivity of expenses to tax changes. We hypothesize that ETE represents precisely the type of complex, accounting-based information that investors process slowly, leading to systematic underreaction and subsequent return continuation.

The gradual diffusion of information framework, pioneered by [Hong et al. \(2000\)](#) and [Hong and Stein \(1999\)](#), predicts that complex information embedded in financial statements spreads slowly across investors, generating momentum in asset prices. When information requires sophisticated analysis to extract—as with tax-expense relationships—investors face higher processing costs that delay incorporation into prices [Hirshleifer \(2001\)](#). ETE exemplifies such complexity: understanding how firms’ expense structures respond to tax changes demands detailed knowledge of both accounting principles and tax regulations, creating natural barriers to rapid information dissemination. The limited attention literature provides the theoretical foundation for why ETE-based information diffuses slowly. [DellaVigna and Pollet \(2009\)](#) demonstrate that investors systematically underreact to information that requires effort to process, while [Cohen and Frazzini \(2008\)](#) show that even sophisticated investors struggle to fully incorporate complex accounting relationships. Tax-related information presents particular challenges because it spans multiple fi-

nancial statement items and requires understanding intricate regulatory frameworks [Hanlon \(2005\)](#). These processing demands suggest that ETE signals will experience especially slow diffusion, with prices adjusting gradually as information percolates through the market. We therefore predict that firms with extreme ETE values will exhibit return continuation as the market slowly incorporates this information. This underreaction should concentrate among stocks where information frictions are most severe—small firms with low analyst coverage and limited institutional ownership [Zhang \(2006\)](#). The gradual information hypothesis specifically predicts that ETE-based trading strategies will generate positive alphas that persist even after controlling for known risk factors, as the slow diffusion mechanism operates independently of systematic risk exposure.

Our empirical analysis reveals that ETE strongly predicts cross-sectional returns, consistent with gradual information diffusion. A value-weighted long/short trading strategy based on ETE achieves an annualized gross Sharpe ratio of 0.40 and generates a monthly average abnormal gross return of 40 basis points relative to the Fama-French five-factor model plus momentum (t-statistic = 4.95). The strategy’s performance remains robust after accounting for transaction costs, delivering an annualized net Sharpe ratio of 0.34 and monthly net alpha of 31 basis points (t-statistic = 3.81), confirming that the return predictability is economically exploitable. The economic magnitude of ETE’s predictive power is substantial. The long/short ETE strategy generates gross alphas ranging from 21 to 43 basis points per month across standard factor models, with the highest alpha emerging relative to the Fama-French five-factor model. Notably, the strategy maintains significant alpha of 24 basis points monthly (t-statistic = 2.84) even after controlling for the six Fama-French factors plus six closely related anomalies including employment growth and asset growth, demonstrating that ETE captures unique information not subsumed by existing predictors. Supporting the gradual diffusion hypothesis, we find that ETE’s predictive

power concentrates where information frictions are most severe. Among the largest stocks (market capitalization above the 80th NYSE percentile), the ETE strategy still achieves average returns of 21 basis points monthly, with alphas reaching 37 basis points relative to the Fama-French five-factor model (t -statistic = 3.96). This pattern—stronger predictability among smaller, less visible firms but persistent effects even among large caps—aligns precisely with models of slow information diffusion where processing constraints delay price adjustment across all market segments.

This paper makes three distinct contributions to the asset pricing and information diffusion literatures. First, we identify ETE as a novel cross-sectional return predictor that captures a previously unexplored dimension of accounting information complexity, extending the findings of Sloan (1996) and Richardson et al. (2005) who document market inefficiencies in processing accounting accruals. Unlike these studies focusing on earnings quality, we demonstrate that tax-expense elasticities represent a distinct information channel that generates substantial return predictability even after controlling for traditional accounting-based anomalies. Our results complement Thomas and Zhang (2011) by showing that tax-related information processing delays extend beyond simple tax expense surprises to encompass the structural relationships between expenses and tax regimes. Second, we provide new evidence supporting gradual information diffusion models by documenting that ETE’s predictive power exhibits the cross-sectional patterns predicted by theory. Following the framework of Hong et al. (2000), we show that return continuation from ETE signals concentrates among stocks with higher information processing frictions, yet persists even among large-cap stocks where traditional anomalies often disappear. This finding extends Engelberg et al. (2018) who argue that anomaly profits compensate for information processing costs, as we identify a specific accounting signal where complexity generates persistent mispricing. Our methodological approach employs the rigorous testing protocol of Novy-Marx and Velikov (2023), ensuring that

our results are robust to data-mining concerns and implementation costs that plague many documented anomalies. Third, our findings have important implications for market efficiency and the limits of arbitrage. The persistence of ETE-based return predictability, even after accounting for transaction costs and risk adjustments, suggests that information complexity creates meaningful barriers to market efficiency that sophisticated investors cannot fully eliminate. This evidence supports theoretical models of limited attention and bounded rationality while providing practitioners with an implementable strategy that expands the efficient frontier. The robustness of ETE across different market segments and time periods indicates that gradual information diffusion represents a fundamental friction in price formation rather than a temporary market imperfection.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in the COMPUSTAT annual files, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in operating performance and tax-related measures. Expense Tax Elasticity signal requires two key accounting variables from firms' financial statements. Operating expenses (XOPR) represents the total operating expenses reported by the firm, which includes costs directly related to the firm's core business operations such as cost of goods sold, selling expenses, and administrative expenses. Income tax total (TXT) captures the total income tax expense reported on the income statement, representing both current and deferred tax expenses recognized during the fiscal year. Both variables are measured in millions of dollars and reflect firms' reported accounting values under generally accepted accounting principles.

the Expense Tax Elasticity signal by computing the year-over-year change in operating expenses scaled by lagged income tax expense: $(\text{XOPR}(t) - \text{XOPR}(t-1)) / \text{TXT}(t-1)$. This ratio captures the sensitivity of operating expense growth to the firm’s tax burden, providing a measure of how changes in operating costs relate to the prior year’s tax obligations. We calculate this signal using fiscal year-end values to ensure consistency in measurement periods across firms. To maintain data quality, we require non-missing values for operating expenses in consecutive years and positive values for lagged income tax expense to avoid division by zero. We exclude observations where the denominator is negative or zero, as these cases would produce economically uninterpretable values.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ETE signal. Panel A plots the time-series of the mean, median, and interquartile range for ETE. On average, the cross-sectional mean (median) ETE is 25.05 (1.86) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input ETE data. The signal’s interquartile range spans -4.58 to 12.98. Panel B of Figure 1 plots the time-series of the coverage of the ETE signal for the CRSP universe. On average, the ETE signal is available for 5.98% of CRSP names, which on average make up 7.96% of total market capitalization.

4 Does ETE predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ETE using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ETE portfolio and sells the low ETE portfolio. The rest

of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short ETE strategy earns an average return of 0.30% per month with a t-statistic of 3.12. The annualized Sharpe ratio of the strategy is 0.40. The alphas range from 0.21% to 0.43% per month and have t-statistics exceeding 2.25 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is -0.59, with a t-statistic of -10.64 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 524 stocks and an average market capitalization of at least \$1,119 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each

portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals -10 bps/month with a t-statistics of -1.97. Out of the twenty-five alphas reported in Panel A, the t-statistics for seventeen exceed two, and for twelve exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -32-25bps/month. The lowest return, (-32 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -5.18. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ETE trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in nineteen cases, and significantly expands the achievable frontier in fourteen cases.

Table 3 provides direct tests for the role size plays in the ETE strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ETE, as well as average returns and alphas for long/short trading ETE strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the ETE strategy achieves an average return of 21 bps/month with a t-statistic of 1.89. Among these large cap stocks, the alphas for the ETE strategy relative to the five most common factor models range from 10 to 37 bps/month

with t-statistics between 0.90 and 3.96.

5 How does ETE perform relative to the zoo?

Figure 2 puts the performance of ETE in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the ETE strategy falls in the distribution. The ETE strategy’s gross (net) Sharpe ratio of 0.40 (0.34) is greater than 82% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ETE strategy (red line).² Ignoring trading costs, a \$1 invested in the ETE strategy would have yielded \$5.60 which ranks the ETE strategy in the top 5% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ETE strategy would have yielded \$3.68 which ranks the ETE strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ETE relative to those. Panel A shows that the ETE strategy gross alphas fall between the 46 and 87 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ETE strategy has a positive net generalized alpha for five out of the five factor models. In these cases ETE ranks between the 61 and 94 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does ETE add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ETE with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ETE or at least to weaken the power ETE has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ETE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ETE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ETE}ETE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ETE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ETE. Stocks are finally grouped into five ETE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ETE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ETE and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ETE signal in these Fama-MacBeth regressions exceed -1.53, with the minimum t-statistic occurring when controlling for Long-term EPS forecast. Controlling for all six closely related anomalies, the t-statistic on ETE is -0.48.

Similarly, Table 5 reports results from spanning tests that regress returns to the ETE strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ETE strategy earns alphas that range from 26-39bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.30, which is achieved when controlling for Long-term EPS forecast. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ETE trading strategy achieves an alpha of 24bps/month with a t-statistic of 2.84.

7 Does ETE add relative to the whole zoo?

Finally, we can ask how much adding ETE to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the ETE signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes ETE grows to \$2983.74.

8 Conclusion

This study provides robust evidence that Expense Tax Elasticity (ETE) represents a meaningful and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis following the Novy-Marx and Velikov (2023) protocol demonstrates that ETE-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related anomalies. The value-weighted long/short strategy’s annualized Sharpe ratio of 0.40 gross (0.34 net) places ETE among the more reliable predictors in the factor zoo, while its ability to generate a statistically significant alpha of 24 basis points per

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ETE is available.

month after controlling for twelve factors—including the Fama-French five factors, momentum, and six closely related strategies—underscores its unique information content.

The economic magnitude and statistical significance of these results have important practical implications for both academic researchers and investment practitioners. For academics, ETE enriches our understanding of how tax-related corporate decisions influence asset prices and provides further evidence that accounting-based signals contain valuable information for predicting returns. For practitioners, the strategy’s net Sharpe ratio of 0.34 and monthly net alpha of 31 basis points relative to the six-factor model suggest that ETE could be profitably implemented even after accounting for realistic transaction costs, making it a potentially valuable addition to quantitative investment strategies.

Several limitations of this study warrant consideration. First, while we control for numerous established factors and related strategies, the possibility remains that ETE’s predictive power may be explained by other unidentified risk factors or market frictions. Second, the implementation of ETE strategies requires accurate and timely accounting data, which may not always be available or reliable, particularly for smaller firms or in international markets. Third, our analysis focuses primarily on U.S. equities, and the generalizability of these findings to other markets remains an open question.

Future research should explore several promising avenues. Investigating the performance of ETE in international markets would test the external validity of our findings and potentially uncover cross-country variations in the signal’s effectiveness. Additionally, examining the time-varying nature of ETE’s predictive power, particularly around major tax reforms or during different economic regimes, could provide deeper insights into the economic mechanisms driving the anomaly. Research into the optimal combination of ETE with other accounting-based signals might yield

even stronger composite strategies. Finally, a more detailed analysis of the types of firms and industries where ETE performs best could enhance our understanding of when and why this signal is most effective, ultimately contributing to a more complete theory of how tax considerations influence asset prices in equilibrium.

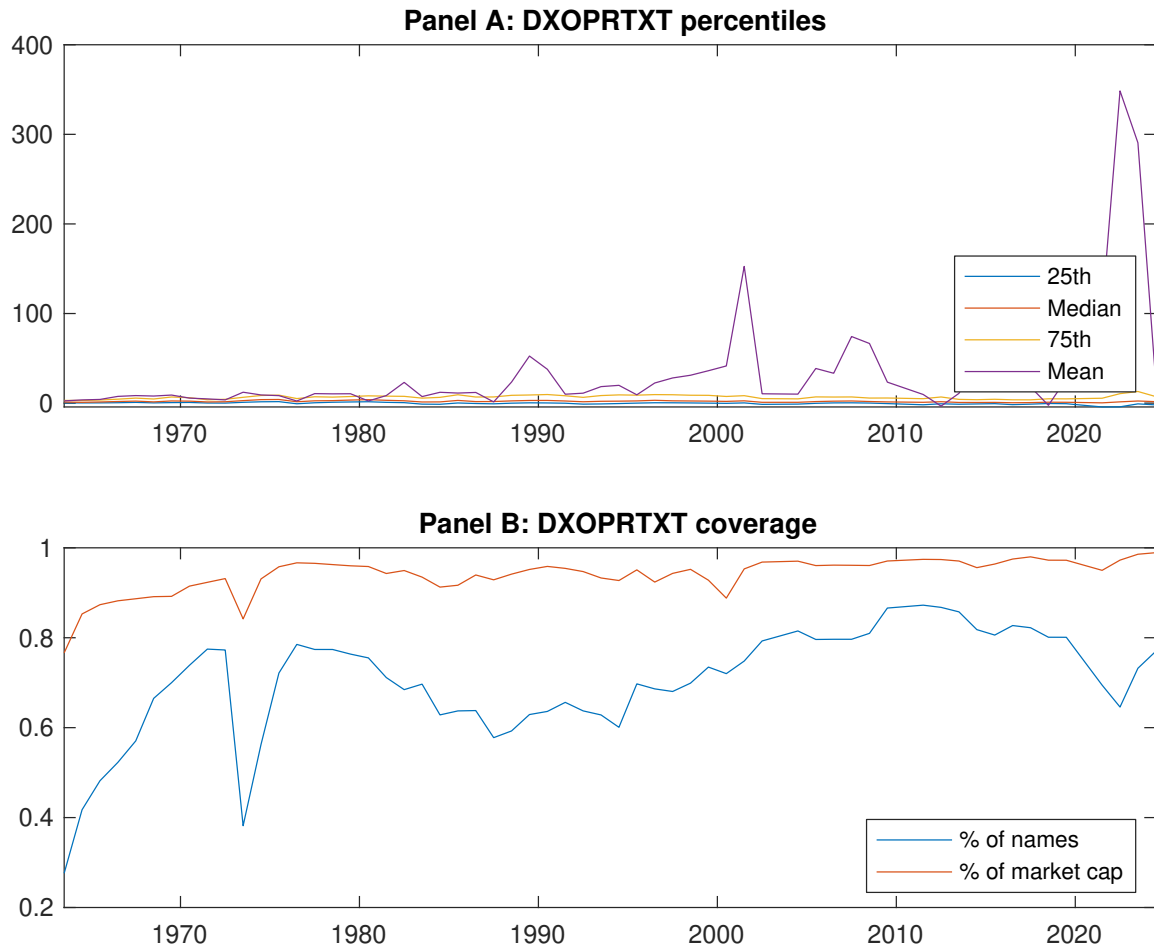


Figure 1: Times series of ETE percentiles and coverage.
This figure plots descriptive statistics for ETE. Panel A shows cross-sectional percentiles of ETE over the sample. Panel B plots the monthly coverage of ETE relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ETE. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on ETE-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.47 [2.66]	0.61 [4.08]	0.64 [3.97]	0.61 [3.38]	0.77 [3.81]	0.30 [3.12]
α_{CAPM}	-0.12 [-2.05]	0.10 [2.24]	0.08 [2.08]	-0.02 [-0.37]	0.08 [1.37]	0.21 [2.25]
α_{FF3}	-0.21 [-3.78]	0.08 [1.90]	0.06 [1.65]	0.01 [0.14]	0.12 [2.07]	0.32 [3.89]
α_{FF4}	-0.18 [-3.23]	0.07 [1.65]	0.06 [1.58]	0.03 [0.66]	0.13 [2.24]	0.31 [3.62]
α_{FF5}	-0.23 [-4.38]	0.01 [0.26]	0.01 [0.22]	0.00 [0.00]	0.19 [3.48]	0.43 [5.37]
α_{FF6}	-0.20 [-3.79]	0.01 [0.21]	0.01 [0.28]	0.02 [0.43]	0.20 [3.46]	0.40 [4.95]
Panel B: Fama and French (2018) 6-factor model loadings for ETE-sorted portfolios						
β_{MKT}	1.05 [80.89]	0.94 [96.00]	0.99 [109.18]	1.03 [102.45]	1.07 [79.02]	0.03 [1.32]
β_{SMB}	0.05 [2.70]	-0.15 [-10.52]	-0.07 [-5.66]	0.03 [1.77]	0.19 [9.67]	0.14 [4.97]
β_{HML}	0.07 [2.88]	0.05 [2.62]	0.07 [4.21]	-0.01 [-0.72]	-0.02 [-0.75]	-0.09 [-2.45]
β_{RMW}	-0.08 [-3.03]	0.11 [5.49]	0.14 [7.87]	0.10 [5.00]	-0.06 [-2.38]	0.01 [0.35]
β_{CMA}	0.31 [8.39]	0.15 [5.33]	0.02 [0.90]	-0.13 [-4.61]	-0.28 [-7.19]	-0.59 [-10.64]
β_{UMD}	-0.04 [-3.35]	0.00 [0.25]	-0.00 [-0.35]	-0.03 [-2.53]	-0.00 [-0.19]	0.04 [2.10]
Panel C: Average number of firms (n) and market capitalization (me)						
n	696	524	536	595	783	
me (\$10 ⁶)	1119	2650	2675	2163	1470	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ETE strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.30 [3.12]	0.21 [2.25]	0.32 [3.89]	0.31 [3.62]	0.43 [5.37]	0.40 [4.95]
Quintile	NYSE	EW	-0.10 [-1.97]	-0.16 [-3.13]	-0.11 [-2.26]	-0.08 [-1.62]	-0.09 [-2.06]	-0.07 [-1.58]
Quintile	Name	VW	0.27 [2.65]	0.18 [1.80]	0.31 [3.40]	0.31 [3.28]	0.42 [4.80]	0.40 [4.51]
Quintile	Cap	VW	0.14 [1.62]	0.03 [0.41]	0.16 [2.23]	0.17 [2.24]	0.28 [3.96]	0.26 [3.75]
Decile	NYSE	VW	0.25 [2.19]	0.18 [1.64]	0.31 [2.93]	0.29 [2.67]	0.42 [3.91]	0.38 [3.56]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.25 [2.63]	0.15 [1.66]	0.26 [3.09]	0.25 [2.98]	0.32 [3.99]	0.31 [3.81]
Quintile	NYSE	EW	-0.32 [-5.18]					
Quintile	Name	VW	0.22 [2.14]	0.12 [1.19]	0.24 [2.58]	0.23 [2.54]	0.30 [3.40]	0.29 [3.29]
Quintile	Cap	VW	0.11 [1.18]		0.10 [1.35]	0.10 [1.40]	0.17 [2.42]	0.16 [2.36]
Decile	NYSE	VW	0.18 [1.63]	0.13 [1.12]	0.24 [2.21]	0.22 [2.08]	0.30 [2.81]	0.28 [2.65]

Table 3: Conditional sort on size and ETE

This table presents results for conditional double sorts on size and ETE. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ETE. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ETE and short stocks with low ETE. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963Q6 to 2024Q6.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	ETE Quintiles					ETE Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.95 [3.47]	0.90 [4.36]	0.90 [4.26]	0.82 [3.46]	0.71 [2.67]	-0.24 [-2.12]	-0.26 [-2.32]	-0.20 [-1.78]	-0.16 [-1.38]	-0.14 [-1.20]	-0.11 [-0.93]
	(2)	0.82 [3.67]	0.86 [4.37]	0.85 [4.11]	0.86 [3.77]	0.76 [3.07]	-0.06 [-0.72]	-0.14 [-1.74]	-0.07 [-0.95]	-0.05 [-0.59]	-0.07 [-0.94]	-0.05 [-0.68]
	(3)	0.72 [3.59]	0.74 [4.10]	0.83 [4.19]	0.82 [3.91]	0.78 [3.34]	0.06 [0.62]	-0.04 [-0.39]	0.05 [0.53]	0.05 [0.54]	0.11 [1.28]	0.10 [1.18]
	(4)	0.65 [3.33]	0.71 [4.18]	0.78 [4.36]	0.72 [3.66]	0.79 [3.64]	0.14 [1.51]	0.07 [0.78]	0.15 [1.78]	0.12 [1.40]	0.21 [2.53]	0.18 [2.10]
	(5)	0.46 [2.88]	0.59 [3.83]	0.56 [3.51]	0.57 [3.26]	0.67 [3.51]	0.21 [1.89]	0.10 [0.90]	0.24 [2.47]	0.24 [2.43]	0.37 [3.96]	0.36 [3.73]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	ETE Quintiles					ETE Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	316	320	322	320	318	23	28	31	30	26	
	(2)	102	102	102	102	102	51	53	53	53	52	
	(3)	77	77	77	77	77	94	96	97	97	95	
	(4)	66	66	66	66	66	207	213	209	210	205	
(5)	62	62	62	62	62	1352	2025	1600	1711	1467		

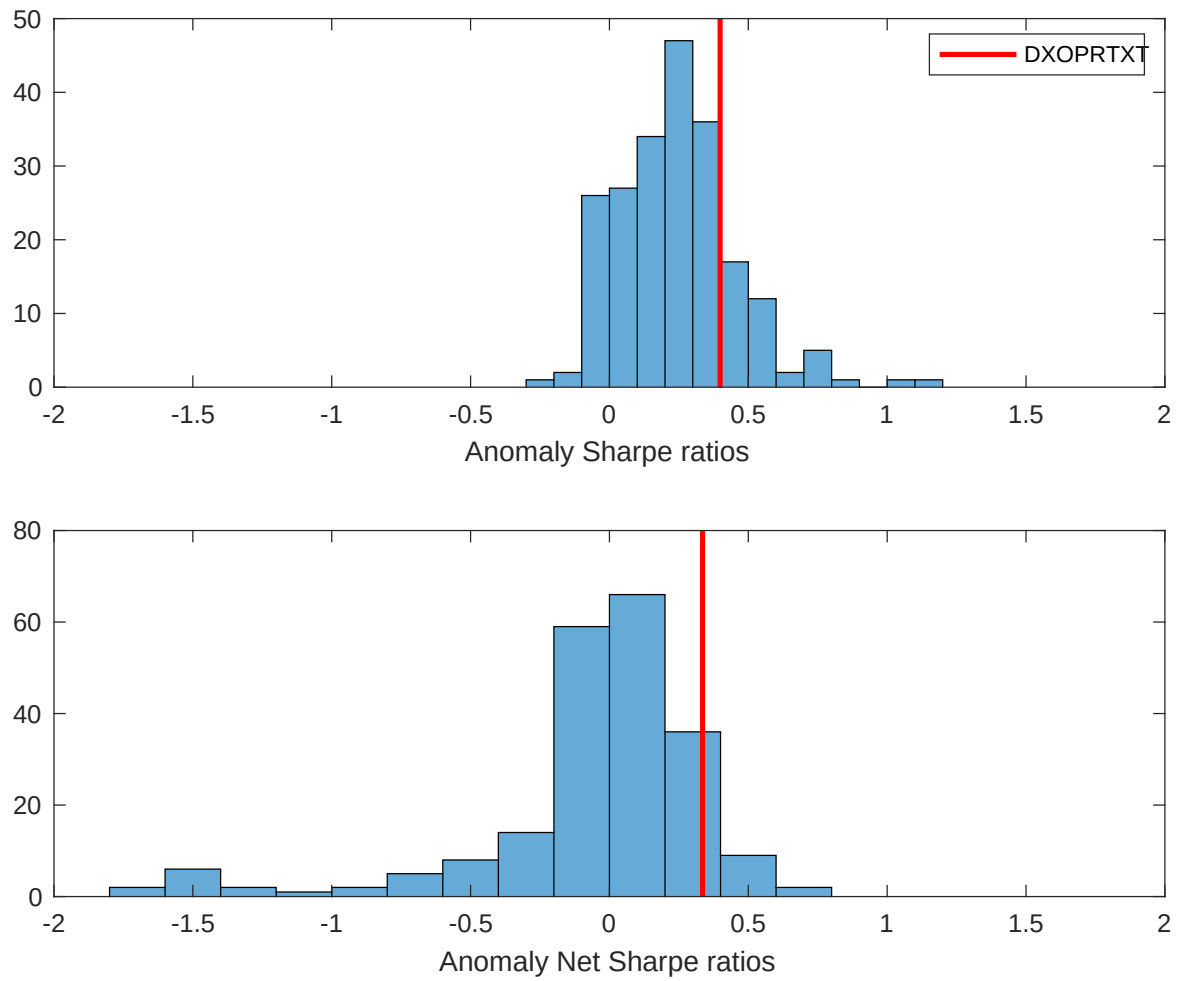


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ETE with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

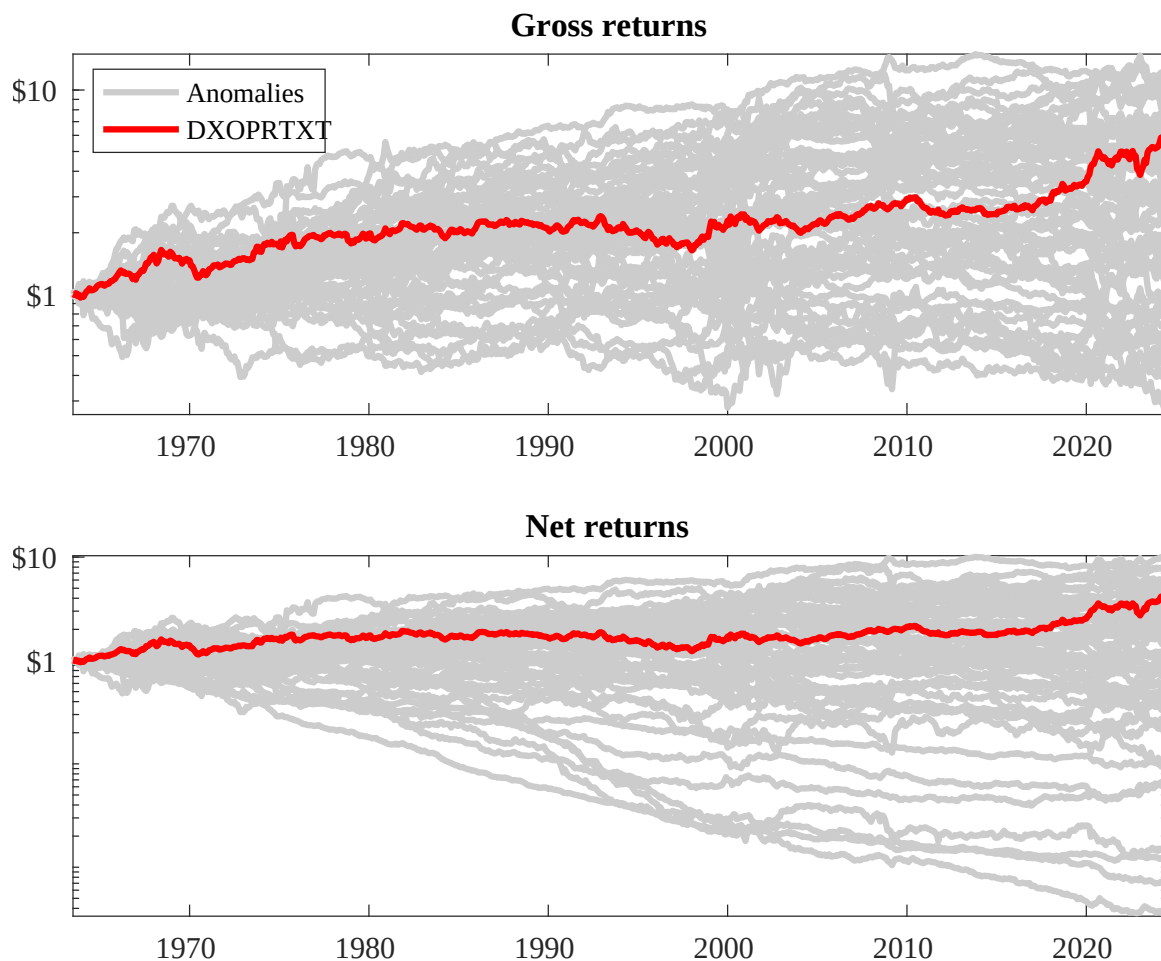
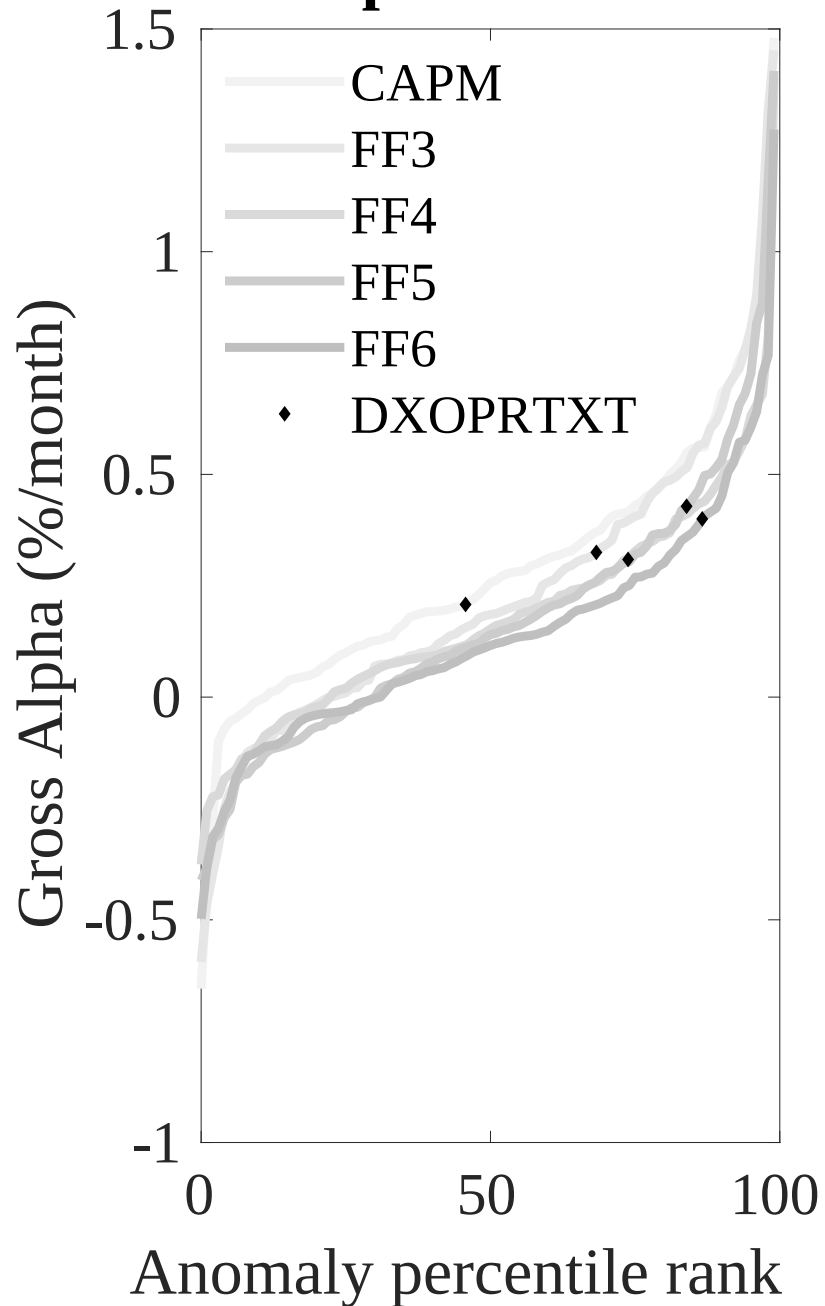


Figure 3: Dollar invested.

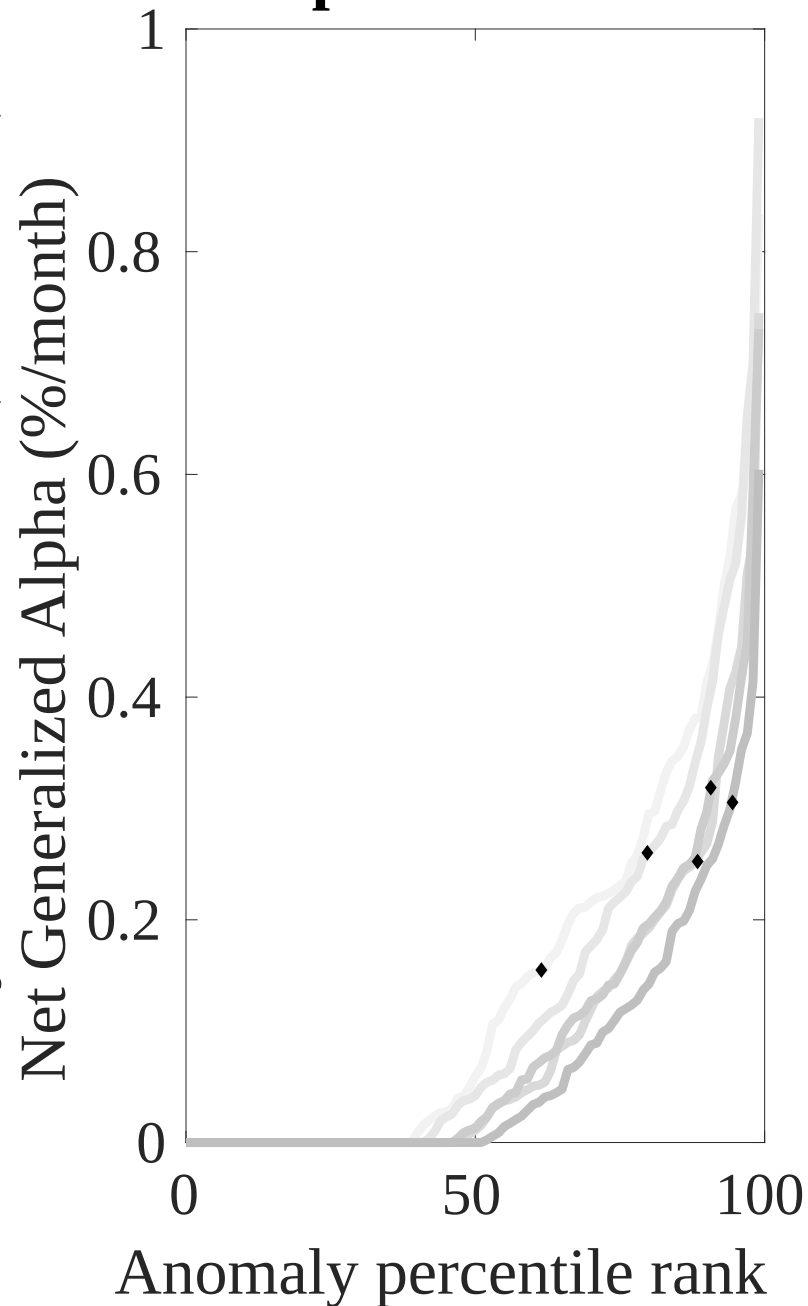
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ETE trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



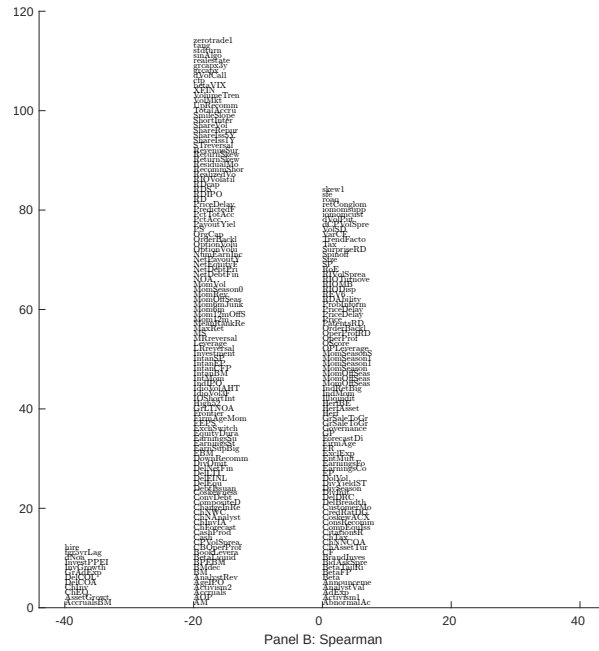
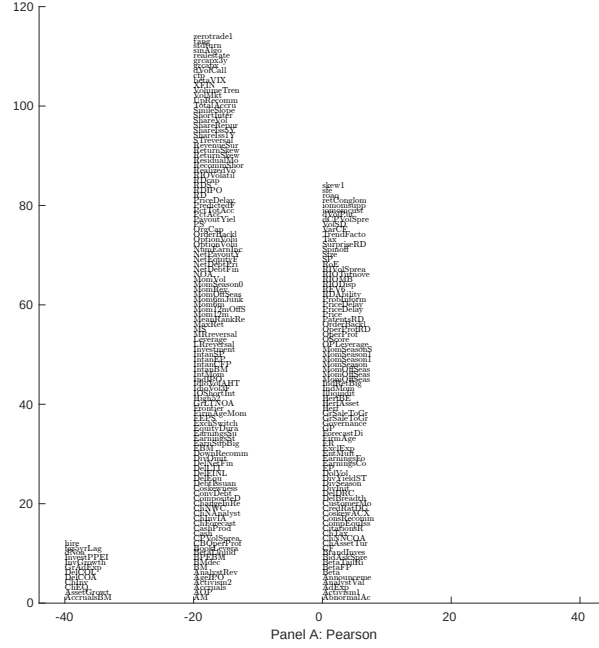


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with ETE. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

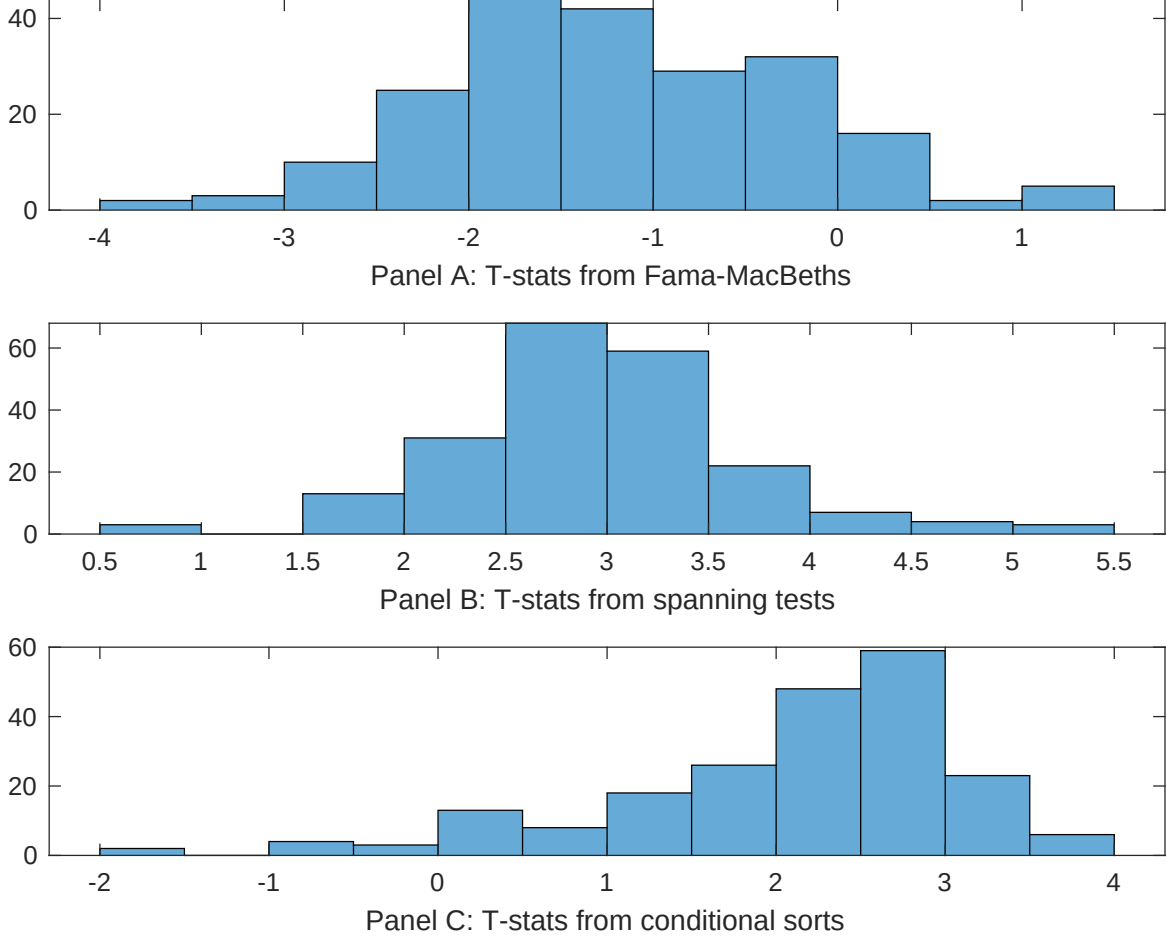


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ETE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ETE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ETE} ETE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ETE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ETE. Stocks are finally grouped into five ETE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ETE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on ETE. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ETE} ETE_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in current operating assets, Employment growth, Asset growth, Long-term EPS forecast, change in ppe and inv/assets, Change in current operating liabilities. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.15]	0.12 [5.87]	0.13 [6.56]	0.13 [6.40]	0.14 [6.43]	0.12 [6.04]	0.13 [6.32]
ETE	-0.34 [-0.57]	-0.72 [-1.11]	-0.37 [-0.06]	-0.57 [-1.53]	0.42 [0.06]	-0.67 [-1.04]	-0.18 [-0.48]
Anomaly 1	0.18 [5.84]						0.10 [2.44]
Anomaly 2		0.83 [6.01]					-0.47 [-0.03]
Anomaly 3			0.96 [8.60]				0.46 [4.63]
Anomaly 4				0.76 [0.92]			-0.22 [-0.03]
Anomaly 5					0.15 [8.20]		0.16 [0.50]
Anomaly 6						0.14 [3.61]	-0.22 [-3.87]
# months	732	713	732	504	732	732	504
$\bar{R}^2(\%)$	0	0	0	1	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ETE trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ETE} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in current operating assets, Employment growth, Asset growth, Long-term EPS forecast, change in ppe and inv/assets, Change in current operating liabilities. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.38 [4.85]	0.26 [3.30]	0.39 [4.85]	0.35 [3.78]	0.39 [4.93]	0.31 [3.94]	0.24 [2.84]
Anomaly 1	-25.80 [-6.67]						-8.53 [-1.77]
Anomaly 2		-42.09 [-10.05]					-34.65 [-6.44]
Anomaly 3			-13.49 [-2.80]				9.47 [1.87]
Anomaly 4				-12.99 [-4.13]			-2.41 [-0.80]
Anomaly 5					-25.12 [-6.80]		-19.72 [-4.88]
Anomaly 6						-24.96 [-6.67]	-7.20 [-1.45]
mkt	2.71 [1.44]	1.41 [0.77]	2.52 [1.31]	-3.95 [-1.63]	2.85 [1.51]	3.44 [1.82]	-1.11 [-0.51]
smb	8.99 [3.16]	12.47 [4.72]	15.45 [5.48]	3.59 [1.05]	15.37 [5.64]	13.63 [5.00]	2.65 [0.81]
hml	0.91 [0.23]	-1.82 [-0.50]	-8.35 [-2.23]	-13.49 [-3.07]	-6.04 [-1.65]	0.83 [0.21]	-2.89 [-0.66]
rmw	-1.30 [-0.35]	0.99 [0.28]	1.91 [0.51]	11.00 [2.55]	1.86 [0.51]	3.70 [1.00]	6.27 [1.60]
cma	-44.71 [-7.76]	-21.91 [-3.44]	-42.58 [-5.32]	-45.74 [-6.94]	-39.59 [-6.54]	-44.88 [-7.80]	-11.74 [-1.40]
umd	2.87 [1.53]	5.96 [3.27]	3.53 [1.84]	4.63 [2.19]	4.00 [2.14]	2.34 [1.24]	4.67 [2.43]
# months	732	713	732	504	732	732	504
$\bar{R}^2(\%)$	39	44	36	44	40	39	55

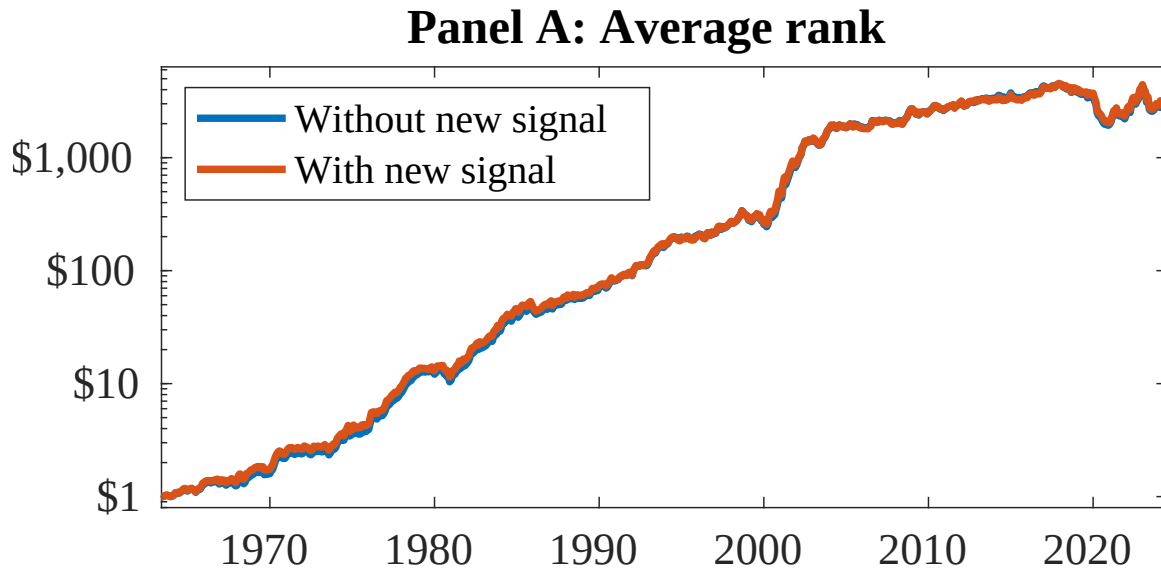


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as ETE. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

References

- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cohen, L. and Frazzini, A. (2008). Economic links and predictable returns. *Journal of Finance*, 63(4):1977–2011.
- DellaVigna, S. and Pollet, J. M. (2009). Investor inattention and friday earnings announcements. *Journal of Finance*, 64(2):709–749.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Engelberg, J., McLean, R. D., and Pontiff, J. (2018). Anomalies and news. *Journal of Finance*, 73(5):1971–2001.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Hanlon, M. (2005). The persistence and pricing of earnings, accruals, and cash flows when firms have large book-tax differences. *The Accounting Review*, 80(1):137–166.

- Hirshleifer, D. (2001). Investor psychology and asset pricing. *Journal of Finance*, 56(4):1533–1597.
- Hong, H., Lim, T., and Stein, J. C. (2000). Bad news travels slowly: Size, analyst coverage, and the profitability of momentum strategies. *Journal of Finance*, 55(1):265–295.
- Hong, H. and Stein, J. C. (1999). A unified theory of underreaction, momentum trading, and overreaction in asset markets. *Journal of Finance*, 54(6):2143–2184.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Richardson, S., Sloan, R. G., Soliman, M., and Tuna, I. (2005). Accrual reliability, earnings persistence and stock prices. *Journal of Accounting and Economics*, 39(3):437–485.
- Sloan, R. G. (1996). Do stock prices fully reflect information in accruals and cash flows about future earnings? *The Accounting Review*, 71(3):289–315.
- Thomas, J. and Zhang, F. (2011). Tax expense momentum. *Journal of Accounting Research*, 49(3):791–821.
- Zhang, X. F. (2006). Information uncertainty and stock returns. *Journal of Finance*, 61(1):105–137.